Name: Dolie Shane B. Aragon

Role in the Group

In this activity, I served as the Client Developer 1. My main responsibility was to develop an application that communicates with the REST API by sending user input and displaying the server's response in a clear and organized format.

Specific Tasks Performed

As the client developer, I created a Python-based client application using the requests library. I designed the program to accept user input, specifically the student ID and subject code, which are required by the API. I implemented the functionality to send HTTP POST requests to the API endpoint and ensured that the response from the server is properly received and displayed. I also added error handling to manage issues such as connection failures and invalid responses. Additionally, I tested the client application multiple times to verify that it communicates correctly with the server.

Code Contributions

My main contribution was the development of a Python script that serves as a client for the REST API. The program prompts the user to enter the required data and stores it in a dictionary, which is then sent to the API using an HTTP POST request. After sending the request, the program attempts to convert the server's response into JSON format using the response.json() method. If successful, the output is formatted using json.dumps() with indentation to make it more readable. If the response is not in JSON format, the raw response text is displayed instead. I also implemented try-except blocks to handle exceptions such as connection errors or request failures, ensuring that the application does not crash and provides meaningful feedback to the user.

Challenges Encountered

During the development of the client application, I encountered several challenges. One of the main difficulties was connecting the Python client to the local server due to incorrect API URLs. I also experienced issues when the server was not running or when XAMPP services were not properly started. Another challenge was related to Python installation, particularly when it was not recognized in the system due to missing PATH configuration. Additionally, I encountered problems when the server response was not in proper JSON format, which required additional handling in the code. Understanding how to properly use POST requests and differentiate them from GET requests was also a challenge at the beginning.

How they contributed to solving problems

To address these challenges, I verified and corrected the API endpoint URL to ensure proper communication between the client and the server. I made sure that XAMPP services, including Apache and MySQL, were running before executing the client program. I also installed Python correctly and ensured that it was added to the system PATH so it could be recognized in the terminal. To handle different types of server responses, I implemented JSON parsing with a fallback to raw text output when necessary. I added exception handling using try-except blocks to catch and display errors clearly. I also coordinated with my group members, especially those responsible for the backend, to ensure that the parameter names and API behavior matched the client implementation. Through these efforts, I was able to successfully develop a working client application.